

Muhammad Fayaz Khan

LLM Engineer | RAG Systems • Agentic AI • FastAPI • LangChain
muhammadfayazkhan50@gmail.com | 03120770266 | Islamabad, Pakistan
github.com/mfayazkhan50-AI | [linkedin.com/in/muhammad-fayaz-khan-271487381](https://www.linkedin.com/in/muhammad-fayaz-khan-271487381)



PROFESSIONAL SUMMARY

LLM Engineer specializing in Enterprise-grade RAG Systems and Agentic AI workflows. Experienced in building production-ready AI applications using FastAPI, LangGraph, LangChain, and Vector Databases with a focus on retrieval accuracy, latency optimization, and LLM cost reduction. Proven ability to design end-to-end AI pipelines, from data ingestion and chunking strategy to deployment and evaluation, with measurable, demoable results.

TECHNICAL SKILLS

Generative AI & RAG	RAG Pipeline, Hybrid Search (Dense + BM25), Neural Reranking, Self-Querying, RAGAS Evaluation
Prompt Engineering	Zero-shot, Few-shot, Chain-of-Thought, ReAct Pattern, Context Engineering
Agentic AI	LangGraph, CrewAI, LangChain, Tool Calling, Multi-Agent Workflows, MCP
Backend & APIs	FastAPI, Python (Async/Await, Decorators, Generators), Pydantic v2, Instructor Library
Automation & Scraping	Playwright, Web Scraping, Scheduled Pipelines
Vector Databases	ChromaDB, Qdrant, Pinecone
Core AI / ML	Machine Learning, Deep Learning (PyTorch), CNNs, Scikit-Learn
DevOps & Tools	Docker, Docker Compose, Git/GitHub, Streamlit, AWS
LLM Providers	OpenAI, Anthropic Claude, Google Gemini, Groq

PROJECTS PORTFOLIO

CompetiTrack AI — Competitor Price & Stock Monitoring Agent [LangGraph · Playwright · FastAPI · Streamlit · Groq]
Built an autonomous agent that scrapes competitor e-commerce websites, detects price and stock changes through a branching LangGraph workflow with error handling, and generates business insights using an LLM. Sends automated email alerts only when a real change is detected, avoiding false-positive noise. Deployed with a FastAPI backend and a Streamlit dashboard for on-demand or scheduled checks.

BizBrain — Business PDF Agent [FastAPI · ChromaDB · RAGAS · RAG · LangChain · Pydantic · Chainlit · Docker]
Enterprise FAQ agent that ingests business PDFs and answers domain-specific queries using Hybrid Search and Cross-Encoder Reranking. Designed for real client deployment with persistent chat history.

Enterprise RAG Platform [FastAPI · ChromaDB · RAGAS · LangChain]
Built a production-grade RAG system using Recursive Semantic Chunking and Cross-Encoder Reranking for querying private PDF data. Achieved high faithfulness and answer relevancy scores via the RAGAS evaluation framework.

Multi-Functional AI Agent [LangChain · Pydantic v2 · Tool Calling]
Developed an autonomous agent with structured Tool Calling and JSON outputs (Pydantic) to handle complex, multi-step business workflows autonomously.

Rossmann Sales Predictor [Python · Scikit-Learn · Feature Engineering]
Predictive model for Rossmann retail chain sales forecasting. Applied advanced feature engineering and ensemble methods to achieve competitive accuracy on time-series retail data.

Churn Predictor [Python · Scikit-Learn · Pandas]
ML classification model to predict customer churn, including data preprocessing, feature selection, and model evaluation with precision/recall analysis.

KEY COMPETENCIES

- Optimizing LLM API costs and latency for production-ready applications

- Implementing LLM caching (GPTCache), reducing API costs by up to 40%
- Building stateful, cyclical multi-agent workflows using LangGraph
- Designing RAG 2.0 pipelines with Hybrid Search and Neural Reranking for enterprise clients
- Building automated web scraping and monitoring pipelines with Playwright and scheduled triggers
- Integrating multiple LLM providers: OpenAI, Anthropic, Gemini, and Groq
- Red-teaming and evaluating LLM outputs with RAGAS and LLM-as-Judge frameworks

EDUCATION

12th Grade (Intermediate Computer Science) — Completed
Self-Taught LLM Engineering Roadmap (2026)